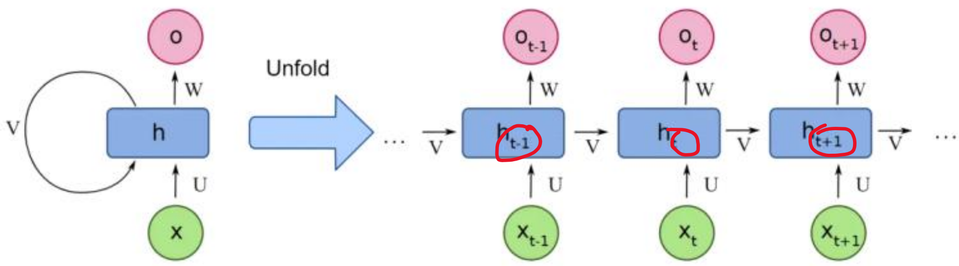
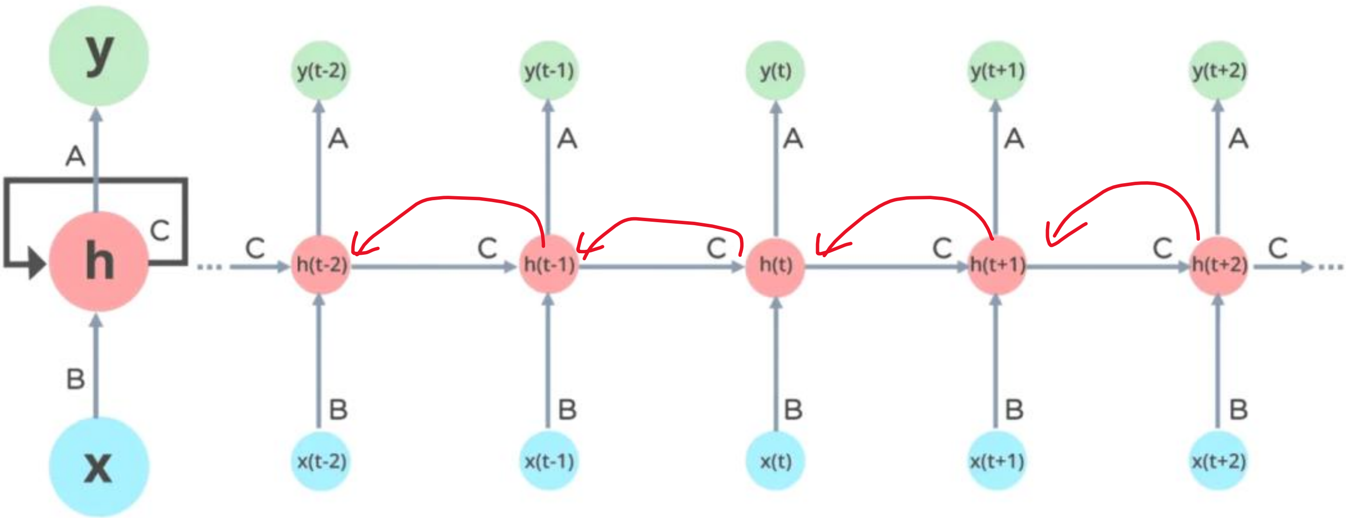
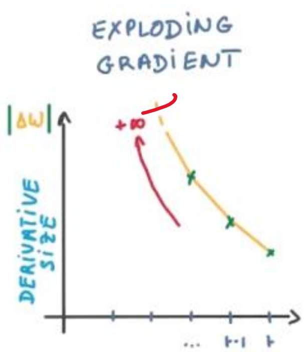
RNN

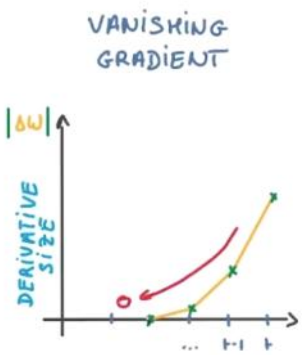
28 March 2026 17:46



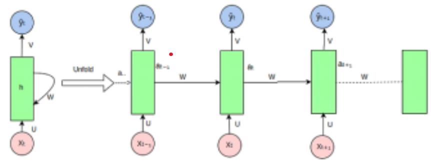


$\frac{\delta e_t}{\delta s_0}$ $\frac{\delta e_t}{\delta s_1}$ e^t





Vanilla
LSTM
gru
x former



$$a_t = f(a_{t-1}, x_t : \underbrace{\theta}_{\{w, b\}})$$

$$a_t = f(u^* x_t + w\, a_{t-1} + b)$$

$$a_4 = f(u\, x_4 + w\, a_3 + b)$$

$$\hat{y} = f(a_t ; \theta)$$

1.

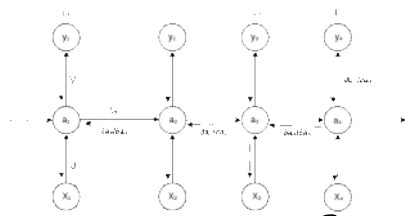
$$a_t = u\, x_t + w\, a_{t-1} + b$$

$$a_t = \tanh(a_t)$$

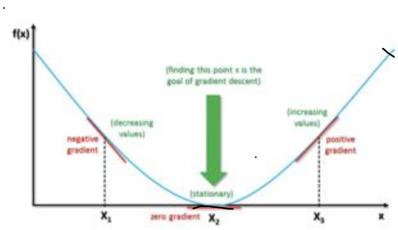
$$\hat{y}_t = \text{softmax}(v\, a_t + c)$$

$$\hat{y} = [\hat{y}_1, \hat{y}_2, \hat{y}_3 \cdots \hat{y}_v]$$

$$L(y, \hat{y}) = \left(\frac{1}{t}\right) \sum (y_t - \hat{y}_t)^2$$



$$\frac{\delta L}{\delta v} = \frac{\delta L}{\delta \hat{y}} \times \frac{\delta \hat{y}}{\delta v}$$



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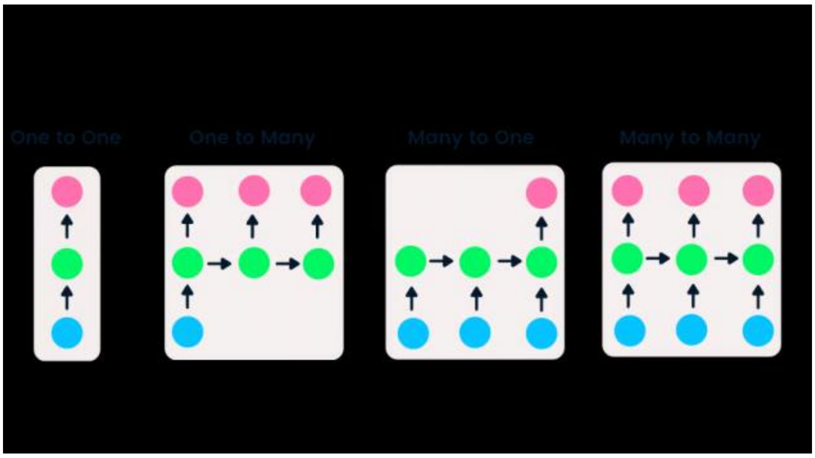
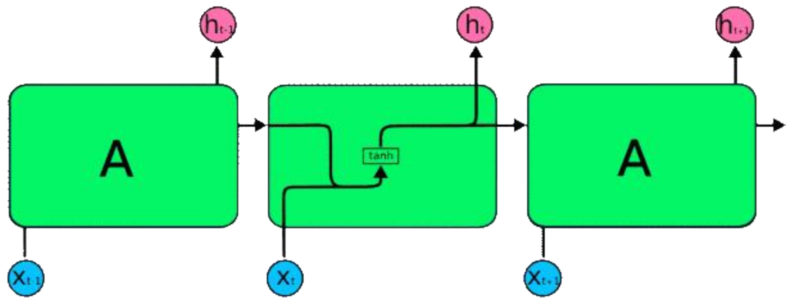




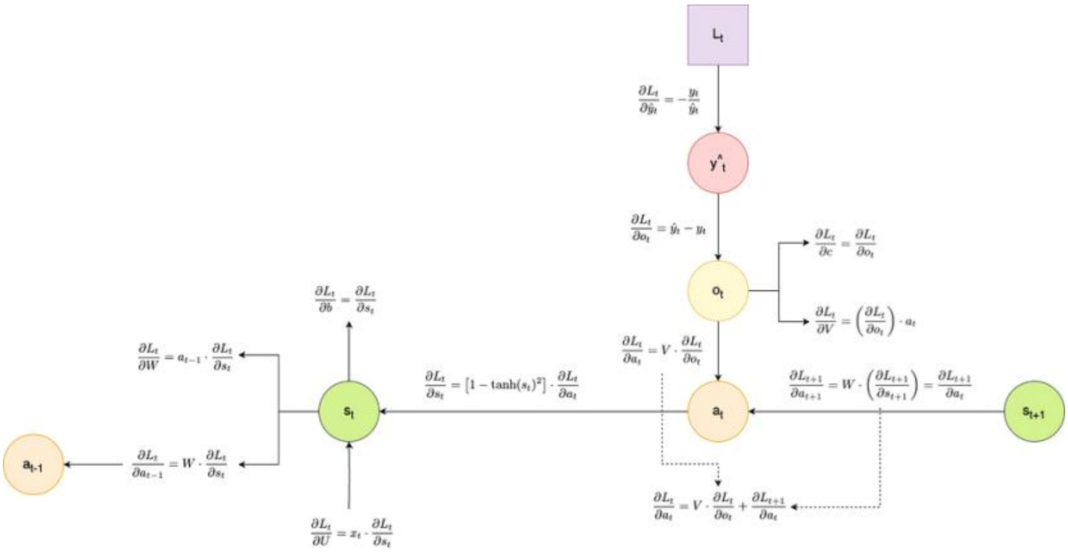
Overall loss function:

$$L = \sum_{t=1}^{T} L_t$$

Loss at each time step:

$$L_t = loss(\hat{y}_t, y_t) = -y_t \log(\hat{y}_t)$$

Where:
L = Overall loss function
L_t = Loss at time step t
$\hat{y}_t$ = Predicted value at time step t
y_t = Actual value at time step t
T = Total number of time steps


L_t
$\frac{\partial L_t}{\partial \hat{y}_t} = -\frac{y_t}{\hat{y}_t}$
$\hat{y}_t$
$\frac{\partial L_t}{\partial o_t} = \hat{y}_t - y_t$
o_t
$\frac{\partial L_t}{\partial c} = \frac{\partial L_t}{\partial o_t}$
$\frac{\partial L_t}{\partial V} = \left(\frac{\partial L_t}{\partial o_t}\right) \cdot a_t$
$\frac{\partial L_t}{\partial a_t} = V \cdot \frac{\partial L_t}{\partial o_t}$
a_t
$\frac{\partial L_{t+1}}{\partial a_{t+1}} = W \cdot \left(\frac{\partial L_{t+1}}{\partial s_{t+1}}\right) = \frac{\partial L_{t+1}}{\partial a_t}$
s_{t+1}
$\frac{\partial L_t}{\partial a_t} = V \cdot \frac{\partial L_t}{\partial o_t} + \frac{\partial L_{t+1}}{\partial a_t}$
$\frac{\partial L_t}{\partial s_t} = \left[1 - \tanh(s_t)^2\right] \cdot \frac{\partial L_t}{\partial a_t}$
s_t
$\frac{\partial L_t}{\partial b} = \frac{\partial L_t}{\partial s_t}$
$\frac{\partial L_t}{\partial W} = a_{t-1} \cdot \frac{\partial L_t}{\partial s_t}$
$\frac{\partial L_t}{\partial a_{t-1}} = W \cdot \frac{\partial L_t}{\partial s_t}$
a_{t-1}
$\frac{\partial L_t}{\partial U} = x_t \cdot \frac{\partial L_t}{\partial s_t}$